Assignment 9.5

Name: kasuri Rohith

Hall no: 2303a510i2

Batch: 23

Problem 1: String Utilities Function

Consider the following Python function:

def reverse_string(text):

return text[::-1]

Task:

1. Write documentation in:

o (a) Docstring

o (b) Inline comments

o (c) Google-style documentation

2. Compare the three documentation styles.

3. Recommend the most suitable style for a utility-based string

library.

Code:

(a) Using Docstring

```python
def reverse_string(text):
    """
    Reverses the given string.

    Args:
        text (str): Input string to reverse.

    Returns:
        str: The reversed string.
    """
    return text[::-1]


if __name__ == "__main__":
    print(reverse_string("Hello"))
```

(b) Inline comments

```python
def reverse_string(text):
    # Function to reverse a given string
    # text: input string
    # Returns: reversed string
    return text[::-1]  # Slicing method to reverse string


# Running the function
print(reverse_string("Hello"))
```

(c) Using Google-Style DocumentaƟon

```python
def reverse_string(text):
    """
    Reverses the given string.
    Args:
        text (str): Input string to reverse.
    Returns:
        str: The reversed string.
    """
    return text[::-1]

# Running the function
print(reverse_string("Hello"))
```

Ouput:

```
Help on module demo1:

NAME
    demo1

FUNCTIONS
    reverse_string(text)
        Reverses the given string.

        Args:
            text (str): Input string to reverse.

        Returns:
            str: The reversed string.

FILE
    c:\users\rohit\links\demo1.py
-- More  --
```

Problem 2: Password Strength Checker

Consider the function:

def check_strength(password):

return len(password) >= 8

Task:

1. Document the function using docstring, inline comments, and

Google style.

2. Compare documentation styles for security-related code.

3. Recommend the most appropriate style.

Code:

(a) Docstring Style

```python
def check_strength(password):
    """
    Checks whether the given password is strong.
    A password is considered strong if it has
    at least 8 characters.

    Parameters:
    password (str): The password string.

    Returns:
    bool: True if strong, False otherwise.
    """
    return len(password) >= 8


# Running
print(check_strength("mypassword"))
print(check_strength("pass"))
```

(b) Inline Comments Style

```python
def check_strength(password):
    """
    Checks whether the given password is strong.
    A password is considered strong if it contains
    at least 8 characters.

    Args:
        password (str): Password provided by the user.

    Returns:
        bool: True if password length is >= 8, else False.
    """
    return len(password) >= 8

# Running
print(check_strength("mypassword"))
print(check_strength("pass"))
```

```python
def check_strength(password):
    """
    Checks whether the given password is strong.
    A password is considered strong if it contains
    at least 8 characters.

    Args:
        password (str): Password provided by the user.

    Returns:
        bool: True if password length is >= 8, else False.
    """
    return len(password) >= 8


# Running
print(check_strength("mypassword"))
print(check_strength("pass"))
```

Output:

```
PS C:\Users\rohit\Links> python -m pydoc demo1
True
False
Help on module demo1:

NAME
    demo1

FUNCTIONS
FUNCTIONS
    check_strength(password)
        Checks whether the given password is strong.
        A password is considered strong if it has
        at least 8 characters.

        Parameters:
        password (str): The password string.

        Returns:
        bool: True if strong, False otherwise.

FILE
    c:\users\rohit\links\demo1.py
-- More  --
```

Problem 3: Math Utilities Module

Task:

1. Create a module math_utils.py with functions:

o square(n)

o cube(n)

o factorial(n)

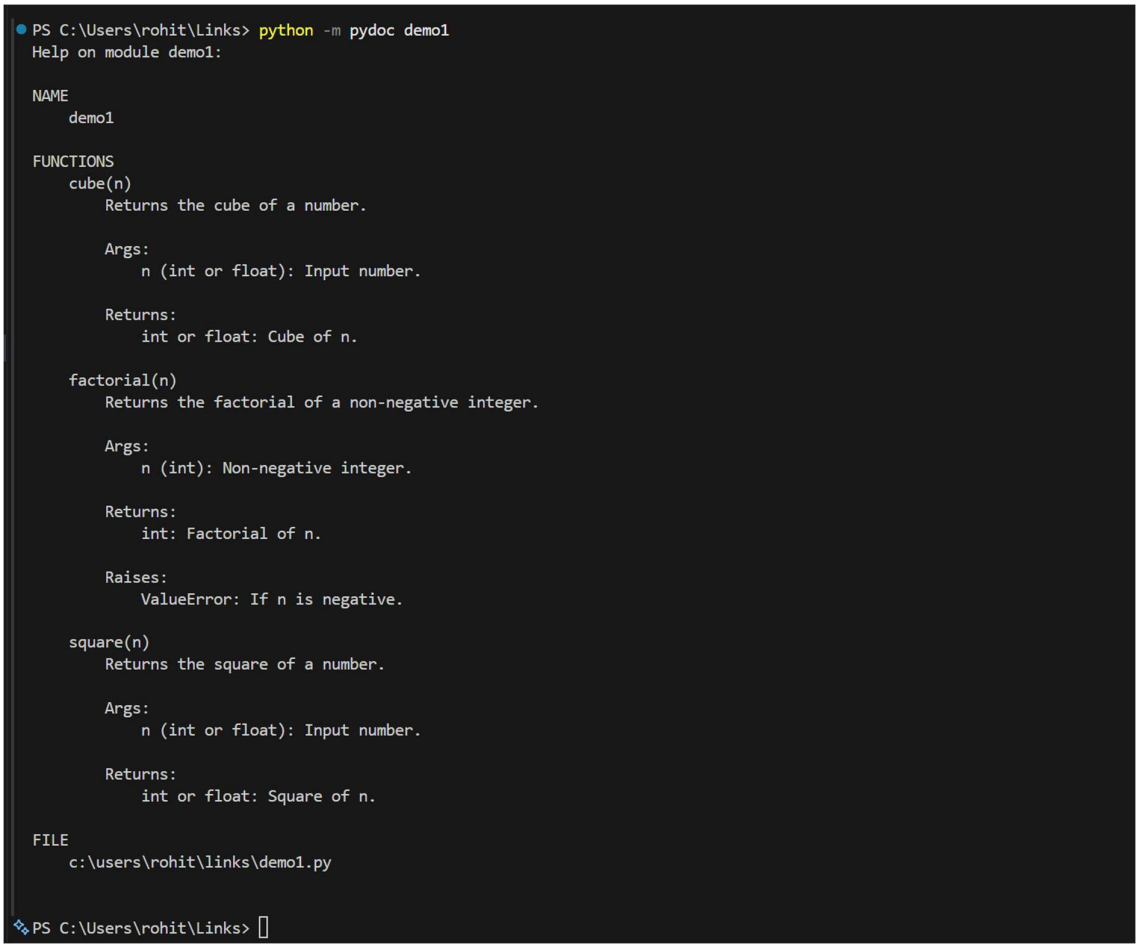
2. Generate docstrings automatically using AI tools.

3. Export documentation as an HTML file.

Code:

```python
def square(n):
    """
    Returns the square of a number.

    Args:
        n (int or float): Input number.

    Returns:
        int or float: Square of n.
    """
    return n * n


def cube(n):
    """
    Returns the cube of a number.

    Args:
        n (int or float): Input number.

    Returns:
        int or float: Cube of n.
    """
    return n * n * n


def factorial(n):
    """
    Returns the factorial of a non-negative integer.

    Args:
        n (int): Non-negative integer.

    Returns:
        int: Factorial of n.

    Raises:
        ValueError: If n is negative.
    """
    if n < 0:
        raise ValueError("Factorial not defined for negative numbers")

    result = 1
    for i in range(1, n + 1):
        result *= i
    return result


if __name__ == "__main__":
    print(square(4))
    print(cube(3))
    print(factorial(5))
```

Output:



Problem 4: Attendance Management Module

Task:

1. Create a module attendance.py with functions:

o mark_present(student)

o mark_absent(student)

o get_attendance(student)

2. Add proper docstrings.

3. Generate and view documentation in terminal and browse

Code:

```python
attendance_record = {}

def mark_present(student):
    """
    Marks a student as present.

    Args:
        student (str): Name of the student.

    Returns:
        None
    """
    attendance_record[student] = "Present"


def mark_absent(student):
    """
    Marks a student as absent.

    Args:
        student (str): Name of the student.

    Returns:
        None
    """
    attendance_record[student] = "Absent"


def get_attendance(student):
    """
    Retrieves the attendance status of a student.

    Args:
        student (str): Name of the student.

    Returns:
        str: Attendance status ('Present', 'Absent', or 'Not Marked').
    """
    return attendance_record.get(student, "Not Marked")


if __name__ == "__main__":
    mark_present("Raju")
    mark_absent("Ramesh")

    print(get_attendance("Raju"))
    print(get_attendance("Ramesh"))
    print(get_attendance("Suresh"))
```

Output:

```
PS C:\Users\rohit\Links> python -m pydoc demo1

NAME
    demo1

FUNCTIONS
    get_attendance(student)
        Retrieves the attendance status of a student.

        Args:
            student (str): Name of the student.

        Returns:
            str: Attendance status ('Present', 'Absent', or 'Not Marked').

    mark_absent(student)
        Marks a student as absent.

        Args:
            student (str): Name of the student.

        Returns:
            None

    mark_present(student)
        Marks a student as present.

        Args:
            student (str): Name of the student.

        Returns:
            None

    mark_present(student)
        Marks a student as present.

        Args:
            student (str): Name of the student.

        Returns:
            None
        Args:
            student (str): Name of the student.

        Returns:
            None
            student (str): Name of the student.

        Returns:
            None

DATA
        Returns:
            None

DATA

DATA
    attendance_record = {}

FILE
    c:\users\rohit\links\demo1.py

FILE
    c:\users\rohit\links\demo1.py


PS C:\Users\rohit\Links>
```

Problem 5: File Handling Function

Consider the function:

def read_file(filename):

with open(filename, 'r') as f:

return f.read()

Task:

1. Write documentation using all three formats.

2. Identify which style best explains exception handling.

3. Justify your recommendation.

```python
def read_file(filename):
    """
    Reads and returns the content of a file.

    Args:
        filename (str): Path to the file.

    Returns:
        str: File contents as a string.

    Raises:
        FileNotFoundError: If the file does not exist.
        OSError: If an OS-related error occurs.
    """
    with open(filename, 'r') as f:
        return f.read()


# Running Example
if __name__ == "__main__":
    print(read_file("sample.txt"))
```

```python
def read_file(filename):
    """
    Reads and returns file content.

    Args:
        filename (str): File name.

    Returns:
        str: File data.

    Raises:
        FileNotFoundError: If file not found.
    """
    with open(filename, 'r') as f:
        return f.read()


if __name__ == "__main__":
    try:
        print(read_file("sample.txt"))
    except Exception as e:
        print("Error:", e)
```

```python
def read_file(filename):
    # Function to read file content
    # filename: name of the file
    # May raise FileNotFoundError if file not found
    with open(filename, 'r') as f:  # open file in read mode
        return f.read()  # return entire content


# Running Example
if __name__ == "__main__":
    print(read_file("sample.txt"))
```

Output:

```
PS C:\Users\rohit\Links> python -m pydoc demo1
Help on module demo1:

NAME
    demo1

FUNCTIONS
    read_file(filename)
        Reads and returns the content of a file.

        Args:
            filename (str): Path to the file.

        Returns:
            str: File contents as a string.

        Raises:
            FileNotFoundError: If the file does not exist.
            OSError: If an OS-related error occurs.

FILE
    c:\users\rohit\links\demo1.py


PS C:\Users\rohit\Links>
```